

Dear Editors,

We would like to submit our manuscript entitled “*A positive-conservative Fourier optimization method for ground states of Bose–Einstein condensates with higher-order interactions*” for consideration in the *Journal of Computational Physics*.

The ground-state problem for Bose–Einstein condensates with higher-order interactions admits a convex formulation in the density variable, but the kinetic term becomes singular near zero density. This creates difficulties for high-accuracy spatial discretization and numerical optimization. In this work, we introduce convex regularizations of the density energy and establish their Γ -convergence as the regularization parameters vanish. The regularized problem is then discretized by a Fourier pseudospectral method that preserves the total mass and enforces nodal positivity.

For the resulting constrained optimization problem, we develop a two-stage solver that combines a potential-proximal accelerated first-order method with a Newton refinement. The proximal treatment avoids placing the large curvature of the regularized potential in the smooth FISTA step, while the Newton stage efficiently reduces the remaining stationarity residual. Numerical experiments demonstrate spectral-type spatial convergence for fixed regularization parameters, quantify the effects of the kinetic and potential regularizations on accuracy and spatial resolution, and illustrate the computational efficiency of the two-stage strategy in one- and two-dimensional examples.

We believe that the manuscript is well suited to the *Journal of Computational Physics*, as it combines variational analysis, structure-preserving Fourier discretization, and constrained optimization for the accurate computation of nonlinear quantum ground states. We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the submission of the manuscript. We also confirm that there are no conflicts of interest to declare.

Thank you very much for your consideration. We look forward to hearing from you.

Sincerely,

Xinran Ruan

School of Mathematical Sciences, Capital Normal University, Beijing, China

Email: xinran.ruan@cnu.edu.cn

On behalf of all authors